

LOSS-INFORMED TRANSFORMER TRAINING WITH LEARNED RATIONAL ACTIVATIONS

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ABSTRACT

We introduce a joint activation–optimizer design that achieves lower mean validation loss than AdamW, Muon, and SOAP in Transformer pretraining at matched training-token budgets. Our approach uses the model’s learned nonlinear responses to guide both the construction of matrix updates and their joint allocation across groups and layers. **GRAIN**, Groupwise Rational Activation with Internal Normalization, learns group-specific rational nonlinearities in normalized feature coordinates and restores the group scale at its output. **TILLER**, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses departures from the initial activation function’s response to rotate matrix update directions. The activation features and their derivatives also provide first-order loss responses at token positions shared across layers. These measurements reveal how the responses of different directions reinforce or offset one another, relationships that their individual average descent scores do not determine. A compact loss ledger retains these relationships as the nonlinearities and directions evolve, providing a history-based measure of a joint update’s size through its loss responses. TILLER uses this measure to balance current-gradient descent against the magnitude of the combined first-order responses across sampled positions. Its allocation redistributes each step’s feed-forward direction norm across groups and layers, allowing individual directions to be amplified, attenuated, or reversed while preserving their combined norm. The gains hold across every tested corpus and model size with both SwiGLU and GRAIN baseline models, demonstrating improvements beyond activation replacement alone.

1 INTRODUCTION

Learning an activation function gives a Transformer control over both its feature representation and the response of those features to weight updates. Fixed nonlinearities such as GELU and the SiLU gate in SwiGLU provide effective transformations (Hendrycks & Gimpel, 2016; Shazeer, 2020). Rational activations make the response shape trainable (Molina et al., 2020), with approximation advantages that motivate their use as expressive neural representations (Boullé et al., 2020; Tang & Townsend, 2026). Group-shared rational functions bring this flexibility to Transformer feed-forward blocks (Yang & Wang, 2025). The changing response also supplies information for optimizing the matrices around it: learning the nonlinearity changes how an update propagates through the network.

The incoming and outgoing projections expose two complementary paths. An incoming update changes features through the activation Jacobian; an outgoing update recombines the activated features. Relating these changes to the downstream loss gives a functional description of each matrix direction. The interaction between directions is equally important. Two groups can have the same average descent signal while their loss responses reinforce or offset one another at shared activation positions. Separate descent scores leave these relationships unresolved. Measuring the responses together makes them available for choosing the strengths of simultaneous updates.

We introduce GRAIN, Groupwise Rational Activation with Internal Normalization, as the learned representation for this construction. Each feature group evaluates its own rational function on RMS-normalized coordinates and multiplies the result by the same normalization scale (Zhang & Sennrich, 2019). The learned coefficients control the response to relative feature magnitudes, while the output retains a dependence on input amplitude. Its Jacobian describes both the local rational slope and the

effect of changing the group scale. The features and their derivatives provide the two response paths used to guide optimization.

Matrix-aware optimization uses matrix structure and activation sensitivity to shape updates (Martens & Grosse, 2015; Vyas et al., 2025; Liu et al., 2025). TILLER, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses the learned responses to construct matrix directions and coordinate their joint effect on the loss. It uses the response geometry to rotate an orthogonalized momentum direction along a factorized adaptive tangent, with orientation selected by the gradient (Shazeer & Stern, 2018). It then records the selected feed-forward directions’ loss responses at shared positions in a compact temporal ledger. Together with the current gradients, this history defines a joint model for allocating update strength across groups and layers. The allocation preserves the combined norm of that step’s feed-forward directions, so the ledger changes where the update acts according to the responses it produces.

This construction gives two training methods. Full TILLER continually adapts directions and their joint allocation throughout training. The staged method applies TILLER for an initial interval, then continues from the learned model with a chosen optimizer. GRAIN remains trainable in both methods. Full TILLER maintains loss-response coordination along the complete trajectory; staged training combines a coordinated training stage with the update rule and computational cost of its continuation optimizer.

Across five pretraining corpora and both evaluated model sizes, GRAIN with full TILLER achieves the lowest mean validation loss among the tested activation–optimizer configurations at matched token budgets. The comparisons include AdamW, Muon, SOAP, and factorized adaptive optimizers. At the larger model size, the evaluated staged configuration also achieves lower mean validation loss than every baseline, while using less total training time than full TILLER on every corpus.

[TODO: Experiment section]

2 RELATED WORK

Padé Activation Units learn rational curves jointly with network weights (Molina et al., 2020). Kolmogorov–Arnold Transformer (KAT) implements group-shared rational functions followed by linear maps; its reported configuration shares numerator coefficients across groups (Yang & Wang, 2025). GRAIN evaluates each group’s rational function in RMS-normalized coordinates (Zhang & Sennrich, 2019) and restores the group scale at its output. It retains group sharing while learning both numerator and denominator coefficients independently for each group.

Approximation theory establishes parameter-efficiency separations between rational and ReLU networks (Boullé et al., 2020). *Rational Neural Networks have Expressivity Advantages*, by Maosen Tang and Alex Townsend, extends these comparisons to bounded-parameter smooth-activation networks and analyzes coefficient redundancy under learned affine normalization (Tang & Townsend, 2026). GRAIN uses an internal normalization–rescaling map with no additional learned affine transform.

Muon orthogonalizes momentum (Jordan et al., 2024), while Adafactor estimates gradient second moments from row and column statistics (Shazeer & Stern, 2018). TILLER combines these direction constructions using the learned activation response. Loss-weighted features and Jacobians determine the extent of a rotation from the polar direction along an adaptive tangent, with its orientation selected by the current gradient.

Joint update models differ in their search directions and the information used to compare them. K-FAC constructs parameter-space Fisher approximations from activation and backpropagated-derivative moments, including cross-layer blocks (Martens & Grosse, 2015). Newton–Muon derives input-second-moment preconditioning from a matrix quadratic surrogate (Du & Su, 2026). Krylov Subspace Descent optimizes the nonlinear objective over a curvature-generated Krylov subspace (Vinyals & Povey, 2012). TILLER places its quadratic model on coefficients multiplying selected pairs of incoming and outgoing group directions. Current gradients supply the linear term. A discounted history of shared-position response probes, augmented between probes by current-gradient information, supplies the quadratic penalty. Its cross terms couple coefficients across groups and layers; the allocation preserves the combined feed-forward direction norm before weight decay.

Optimizer switching combines update rules along a training trajectory. SWATS selects an Adam-to-SGD transition using a projection-based trigger (Keskar & Socher, 2017). Our staged method applies TILLER until a chosen step, then continues with a selected optimizer while retaining the learned parameters and trainable GRAIN activations.

3 METHOD

We introduce GRAIN in Transformer feed-forward blocks and TILLER to coordinate matrix updates using GRAIN’s loss response. We first define the block, then the update shared by full-horizon and staged training.

3.1 GRAIN

GRAIN (Groupwise Rational Activation with Internal Normalization) replaces a fixed nonlinearity with group-shared rational functions. For input $x \in \mathbb{R}^d$, partition the intermediate features into G groups of width w . The incoming and outgoing projections of group g are $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. Its feed-forward block is

$$\begin{aligned} z_g &= A_g x, & \rho_g &= \sqrt{\|z_g\|_2^2/w + \epsilon_{\text{act}}}, \\ h_g &= \rho_g \phi_g(z_g/\rho_g), & y &= \sum_{g=1}^G B_g h_g. \end{aligned} \quad (1)$$

The curve acts coordinatewise on RMS-normalized features and its output is multiplied by the same group scale (Zhang & Sennrich, 2019). The scale is computed per token and group, with stabilizer $\epsilon_{\text{act}} > 0$.

Each group learns a rational curve (Molina et al., 2020; Yang & Wang, 2025):

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

The denominator is at least one, so the curve has no poles. All coefficients are shared across channels and token positions within a group and learned independently across groups. We initialize each curve with the same rational approximation to SiLU and train its coefficients jointly with the projection weights. Appendix A derives the Jacobian of the normalized activation and its dependence on input scale.

3.2 TILLER

TILLER (Tangent-Informed Loss Ledger for Equal-budget Reweighting) redistributes a fixed update norm using current descent scores and the joint loss responses of matrix directions. For an L -layer model with G groups per layer, let $i = (\ell, g)$ index the $C = LG$ pairs. Given directions (q_i^A, q_i^B) , constructed below, coefficient c_i multiplies both members of pair i . Define their current descent scores $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ and squared norms $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$, where g_i^A, g_i^B are the accumulated, averaged, clipped gradients. Let G_t be the positive semidefinite metric of accumulated directional loss responses and r_t its cross term with weight decay. With learning rate $\eta_t > 0$, we seek coefficients that minimize the response model

$$\begin{aligned} F_t(c) &= -\eta_t s_t^\top c + \frac{\eta_t^2}{2} (c^\top G_t c + 2r_t^\top c), \\ &\sum_i \omega_i c_i^2 = \sum_i \omega_i. \end{aligned} \quad (3)$$

The linear term rewards descent predicted by the current gradients, while the quadratic term penalizes the combined response of the selected directions and weight decay. The constraint fixes the total feed-forward direction norm at that of the reference allocation $c = \mathbf{1}$, before decay.

For positive group energies, write $W = \text{Diag}(\omega)$. The stationary equation is $(G_t + \lambda_c W)c = s_t/\eta_t - r_t$. We seek a coefficient proposal on its positive-multiplier branch in a Krylov subspace. A

finite proposal is accepted if it reduces F_t relative to $\mathbf{1}$, satisfies the norm tolerance, and meets the layerwise gradient and momentum contraction conditions. Otherwise the unit allocation is retained. Appendix B derives the projected stationarity equation and states the acceptance conditions.

We measure directional loss responses at K example-token positions shared across layers. For block input $x_{k\ell}$, activated features h_{ki} , and activation Jacobian $J_{ki} = \partial h_{ki} / \partial z_{ki}$, the paired response is

$$S_{t,ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle, \quad (4)$$

where $\xi_{k\ell}$ is the derivative of the summed minibatch loss at the block output, multiplied by the gradient-clipping factor. The vector $S_{t,c}$ contains the combined sampled loss responses. Appendix C derives this differential and the quadratic response model.

The ledger retains a discounted second moment of the response rows. Periodic probes measure positional variation; between probes, current gradient contractions supply repeated rows at the latest measured response scale. A Robust Frequent Directions sketch stores a row factor and an isotropic remainder (Ghashami et al., 2016; Luo et al., 2019). The current metric includes new rows before compression. Appendix C.1 gives the metric’s spectral approximation and bounds its effect on the coefficient model.

For either matrix in a pair, a calibrated polar direction p and a factorized adaptive direction a transform the same momentum (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). Let $a^\perp$ be the component of a orthogonal to p . TILLER uses loss-weighted summaries of the GRAIN response to set the rotation: participation π measures spread within the group’s Jacobian or activated features, and congruence κ measures alignment with the initial response kernels at the same current inputs. An unchanged kernel retains the polar direction; departure from it permits a rotation whose magnitude increases with participation. The candidate is

$$\tilde{q} = \sqrt{1-e} p + \sigma \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} a^\perp, \quad e = \pi(1-\kappa^2). \quad (5)$$

The sign σ makes the tangent’s gradient contraction nonnegative. A finite, nondegenerate candidate is selected when both its gradient contraction and the parent’s are positive; otherwise p is retained. Matrix-size scaling of the selection gives q . Appendix D defines these quantities and derives the rotation’s norm and angle identities.

3.2.1 FULL-HORIZON TILLER

Full-horizon training recomputes directions and allocations at every update. With decay rate λ_{wd} , it applies $A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}}) A_i - \eta_t c_i q_i^A$ and the corresponding update to B_i . Attention matrices use response-guided rotations with layerwise GRAIN summaries and their own moments; their directions are applied directly. Rational coefficients and remaining parameters use AdamW (Loshchilov & Hutter, 2019).

3.2.2 STAGED TRAINING

For training horizon T and transition $T_s < T$, staged training uses TILLER through update T_s and a selected continuation optimizer $\mathcal{U}$ afterward. Its state initializer $\mathcal{I}$ carries over first-moment histories compatible with the continuation’s recurrence and matrix grouping. Additional statistics use the initialization of $\mathcal{U}$. The first continuation update acts on the learned feed-forward and attention matrices at $T_s + 1$.

Model parameters and the learning-rate schedule continue across the transition. GRAIN’s rational coefficients remain trainable and retain their AdamW state. Response collection and ledger updates end at T_s . Table 1 presents the training loop for both forms.

4 EXPERIMENTS

[TODO: Experiment section]

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with method development, experimental planning, implementation, mathematical derivations and proof checking, literature analysis, result interpretation, and manuscript drafting and revision.

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A DIFFERENTIAL PROPERTIES OF GRAIN

For one feature group, write $\Phi(z) = \rho\phi(u)$, where $u = z/\rho$ and $\rho = (z^\top z/w + \epsilon_{\text{act}})^{1/2}$. The coefficients of ϕ are fixed when taking derivatives with respect to z . Differentiating the scale and normalized input gives

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1}(I - uu^\top/w). \quad (6)$$

Applying the product rule to $\rho\phi(u)$ then yields

$$J = \text{Diag}(\phi'(u)) + \frac{[\phi(u) - u \odot \phi'(u)]u^\top}{w}. \quad (7)$$

The first term is the coordinatewise rational slope. The second is the response through the shared normalization scale. At the absolute-value kinks, we use the selected derivative $\text{sign}(0) = 0$.

This expression distinguishes changes in feature shape from changes in amplitude. A perturbation orthogonal to u leaves ρ unchanged to first order, so only the diagonal term acts. For a radial perturbation, set $r_\phi = \phi(u) - u \odot \phi'(u)$ and use $\|u\|_2^2/w = 1 - \epsilon_{\text{act}}/\rho^2$ to obtain

$$Jz = \Phi(z) - \frac{\epsilon_{\text{act}}}{\rho} r_\phi. \quad (8)$$

In particular, when $\epsilon_{\text{act}} = 0$ and $z \neq 0$, $\Phi(az) = a\Phi(z)$ for every $a > 0$. The rational curve changes the response shape while positive input rescaling is carried to the output. Equation 8 gives the correction for a positive stabilizer. For a linear curve $\phi(u) = au$, $r_\phi = 0$ and $\Phi(z) = az$ for every stabilizer value.

The two derivative actions used in TILLER follow from the same diagonal-plus-rank-one form:

$$Jv = \phi'(u) \odot v + r_\phi \frac{u^\top v}{w}, \quad J^\top \zeta = \phi'(u) \odot \zeta + u \frac{r_\phi^\top \zeta}{w}. \quad (9)$$

Thus both the forward response to a perturbation and the backward loss sensitivity retain the effect of normalization. Each action uses one inner product in addition to coordinatewise products.

B THE FIXED-NORM ALLOCATION PROBLEM

Consider prescribed paired directions (q_i^A, q_i^B) . Their parameter blocks are disjoint, so the feed-forward update before weight decay has norm

$$\|\Delta\theta(c)\|_2^2 = \eta_t^2 \sum_i c_i^2 \left(\|q_i^A\|_F^2 + \|q_i^B\|_F^2 \right) = \eta_t^2 c^\top Wc, \quad W = \text{Diag}(w). \quad (10)$$

The constraint $c^\top Wc = \mathcal{E}_0$, where $\mathcal{E}_0 = \mathbf{1}^\top W\mathbf{1}$, therefore fixes the direction norm at its unit-allocation value. A shared coefficient preserves the relative scaling of each group's incoming and outgoing directions, including when that coefficient is negative.

For positive ω_i and η_t , substitute $y = W^{1/2}c$ in the objective. This gives

$$\frac{F_t(c)}{\eta_t^2} = \mathcal{Q}_t(y) = \frac{1}{2}y^\top M_t y - b_t^\top y, \quad \|y\|_2^2 = \mathcal{E}_0, \quad (11)$$

$$M_t = W^{-1/2}G_t W^{-1/2}, \quad b_t = W^{-1/2}(s_t/\eta_t - r_t).$$

Differentiating the Lagrangian $\mathcal{Q}_t(y) + \lambda(\|y\|_2^2 - \mathcal{E}_0)/2$ yields $(M_t + \lambda I)y = b_t$, the relation stated in the Method. The positive semidefiniteness of G_t makes $M_t + \lambda I$ positive definite for $\lambda > 0$.

TILLER represents this relation in a Lanczos-generated Krylov subspace (Lanczos, 1950). Let Q_k be its orthonormal basis and let T_k be the Lanczos tridiagonal model. Clipping negative numerical eigenvalues gives $T_k^+ = U \text{Diag}(\theta)U^\top$. For $\gamma = U^\top Q_k^\top b_t$, projected stationarity gives

$$y = Q_k U \text{Diag}((\theta + \lambda)^{-1})\gamma, \quad \psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (12)$$

The scalar equation enforces the norm of the projected point. Its positive branch has the following characterization.

Proposition 1 (Positive-multiplier allocation). *Let $T_k^+ \succeq 0$, $Q_k^\top Q_k = I$, and $\gamma \neq 0$. Equation 12 has at most one root with $\lambda > 0$. If such a root exists, its projected coordinates are the unique minimizer of the clipped projected quadratic on the sphere of squared radius $\mathcal{E}_0$.*

Proof. For positive λ , $\psi'(\lambda) = -2 \sum_j \gamma_j^2 / (\theta_j + \lambda)^3 < 0$, which proves uniqueness of the root. Put $b_k = Q_k^\top b_t$ and $z_* = (T_k^+ + \lambda I)^{-1} b_k$. For any z on the same sphere, stationarity and equality of norms give

$$\left(\frac{1}{2} z^\top T_k^+ z - b_k^\top z\right) - \left(\frac{1}{2} z_*^\top T_k^+ z_* - b_k^\top z_*\right) = \frac{1}{2} (z - z_*)^\top (T_k^+ + \lambda I) (z - z_*). \quad (13)$$

The right side is positive unless $z = z_*$. $\square$

The mapped coefficients $c = W^{-1/2} y$ supply a proposal for F_t . Its improvement is evaluated with the current metric G_t ; acceptance also requires the prescribed norm tolerance.

Norm preservation permits sign changes within a layer. Their aggregate effect on a layer and matrix role is measured by

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle, \quad r \in \{A, B\}. \quad (14)$$

Requiring these contractions to be positive for both the proposal and the unit allocation constrains the whole incoming and outgoing updates, while allowing individual group coefficients to be negative.

C THE JOINT LOSS-RESPONSE MODEL

Consider the block $B_i \Phi_i(A_i x_{k\ell})$ at a fixed position. The derivative along its paired matrix direction is

$$\left. \frac{d}{d\alpha} (B_i + \alpha q_i^B) \Phi_i((A_i + \alpha q_i^A) x_{k\ell}) \right|_{\alpha=0} = B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki}. \quad (15)$$

Contracting with the downstream loss derivative gives $S_{t,ki}$ in equation 4. The derivative at a hidden position collects its effect on the entire downstream loss. With microbatch mean loss over N positions and M accumulation steps, the captured derivative $\bar{\xi}$ has scale $1/(NM)$; thus $\xi = NM \chi_{\text{clip}} \bar{\xi}$, where χ_{clip} is the realized gradient-clipping factor, held fixed in this calculation. Averaging over all positions and processes recovers the clipped directional derivative $s_i = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$.

The full gradient determines average descent. Shared-position measurements additionally determine how directions act together. For columns S_i and S_j , their contribution to the squared joint response is $2c_i c_j S_i^\top S_j / K$. For the scaled directions, the sign of $c_i c_j S_i^\top S_j$ determines reinforcement or cancellation. These cross terms couple groups even when they occupy disjoint parameter blocks. Weight decay has the same differential: replacing the directions by $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ and summing over i gives a response a^{dec} . Expanding the combined response,

$$\frac{\|Sc + a^{\text{dec}}\|_2^2}{K} = c^\top \frac{S^\top S}{K} c + 2c^\top \frac{S^\top a^{\text{dec}}}{K} + \frac{\|a^{\text{dec}}\|_2^2}{K}. \quad (16)$$

The final term is independent of the allocation. The first two are the response metric and decay cross term in F_t . Their discounted history supplies the quadratic penalty, while the current s_t supplies the linear descent term.

C.1 APPROXIMATION OF THE RESPONSE HISTORY

Let S_t denote the measured rows on a probe step. Between probes, the row model is $S_t = \mathbf{1}_K (\gamma_t s_t)^\top$, where $\gamma_t = \nu / \|s_t\|_2$ for nonzero s_t and $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ is the latest measured scale. This model has $S_t^\top S_t / K = \nu^2 s_t s_t^\top / \|s_t\|_2^2$: its orientation follows the current gradient contractions and its trace equals the measured ν^2 . For $s_t = 0$, set $\gamma_t = 0$, giving zero rows. The corresponding decay rows are $a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}$, with d_t^{dec} the gradient contraction with the decay direction.

For the resulting sequence of measured and modeled responses, with the second-moment decay $\beta = \beta_2$, define

$$G_t^* = \beta G_{t-1}^* + (1 - \beta) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (17)$$

Each column retains its layer-group identity; each historical row records the direction used at its own training step. The ledger approximates this metric by $\hat{G}_t = \mathcal{B}_t^\top \mathcal{B}_t / K + \tau_t I$. If $\mathcal{A}_t = [\sqrt{\beta} \mathcal{B}_{t-1}; \sqrt{1-\beta} S_t]$, the metric for the current allocation is

$$G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \beta \tau_{t-1} I, \quad r_t = \beta r_{t-1} + (1-\beta) S_t^\top a_t^{\text{dec}} / K. \quad (18)$$

The first observation is unattenuated: $\mathcal{A}_1 = \mathcal{B}_1 = S_1$, $\tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$. Fresh rows enter G_t before compression.

The Robust Frequent Directions approximation (Ghashami et al., 2016; Luo et al., 2019) has a direct spectral description. Let σ_j and v_j be the singular values and right singular vectors of $\mathcal{A}_t$. For a retained rank r , set $\Delta_t = \sigma_{r+1}^2$, or zero if that singular value is absent. The retained factor has rows $\sqrt{\sigma_j^2 - \Delta_t} v_j^\top$ for the available $j \leq r$, and $\tau_t = \beta \tau_{t-1} + \Delta_t / (2K)$. The isotropic increment centers the error from spectral shrinkage.

Proposition 2 (Error in the response metric). *For finite response rows, $K > 0$, $r \geq K$, $0 \leq \beta \leq 1$, and $\tau_{t-1} \geq 0$, $\hat{G}_t$ is positive semidefinite and*

$$\|\hat{G}_t - G_t\|_2 \leq \frac{\Delta_t}{2K}, \quad |c^\top (\hat{G}_t - G_t) c| \leq \frac{\Delta_t}{2K} \|c\|_2^2. \quad (19)$$

Proof. The factor Gram matrix and isotropic term are positive semidefinite. In the right-singular-vector basis, a retained direction loses Δ_t/K by shrinkage and gains $\Delta_t/(2K)$, giving error $-\Delta_t/(2K)$. A discarded direction with squared singular value $s \in [0, \Delta_t]$ has error $(\Delta_t/2 - s)/K$. The same expression with $s = 0$ covers null directions. Every error eigenvalue therefore lies in $[-\Delta_t/(2K), \Delta_t/(2K)]$, proving both bounds. $\square$

Without the isotropic increment, the error interval is $[-\Delta_t/K, 0]$. Centering halves this one-compression spectral bound. Let $E_t = \hat{G}_t - G_t^*$. Since $E_t = \beta E_{t-1} + (\hat{G}_t - G_t)$ and $E_1 = 0$,

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta^{t-j} \frac{\Delta_j}{2K}, \quad G_t - G_t^* = \beta E_{t-1}. \quad (20)$$

This connects the sketch approximation to the coefficient model. If $\|G_t - G_t^*\|_2 \leq \varepsilon_t$, replacing G_t by G_t^* changes $F_t(c)$ by at most $\eta_t^2 \varepsilon_t \|c\|_2^2 / 2$. Under the fixed-norm constraint and positive ω_i , this is at most $\eta_t^2 \varepsilon_t \mathcal{E}_0 / (2 \min_i \omega_i)$, where $\mathcal{E}_0 = \sum_i \omega_i$.

D GEOMETRY OF RESPONSE-GUIDED DIRECTIONS

The factorized direction is determined by the marginals of the full gradient second-moment history. Let R_t and C_t denote its row and column sums, and let m_t be the shared momentum. Since summation commutes with exponential averaging, these marginals can be represented by their own histories. With zero-initialized moments, the adaptive transformation is

$$V_t = \frac{R_t C_t^\top}{(\mathbf{1}^\top R_t)(1 - \beta_2^t)}, \quad a_t = \frac{m_t}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (21)$$

All operations in the second expression are entrywise. For positive total second moment, the normalized outer product is the rank-one approximation determined by its row and column marginals (Shazeer & Stern, 2018).

The group calibration follows from the norm of an exact polar factor. The parent is $p = \chi \mathcal{P}_5(m)$, where $\mathcal{P}_5$ is the five-iteration Newton–Schulz polar approximation (Jordan et al., 2024; Liu et al., 2025). For G groups of width w and total intermediate width $H_r = Gw$, $\chi = \sqrt{\min(H_r, d) / (G \min(w, d))}$. An exact full-rank polar factor of a group matrix has squared Frobenius norm $\min(w, d)$; hence this calibration makes the sum over groups equal the whole-projection value $\min(H_r, d)$. The selected feed-forward direction also receives the usual whole-matrix multiplier $\mu = 0.2 \sqrt{\max(H_r, d)}$.

The rotation is specified by the learned response geometry. For nonzero J and h , define

$$\text{pr}_A(J) = \frac{\text{tr}(JJ^\top)^2}{w\|JJ^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w\|h\|_4^4}. \quad (22)$$

Both are normalized participation ratios. The first applies to squared singular values of the Jacobian and the second to squared feature coordinates. Equal energy in m nonzero components gives m/w ; the ratios lie in $[1/w, 1]$ by Cauchy–Schwarz. Their weighted averages, denoted π_i^A and π_i^B , use $\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2/w$ and $\alpha_{ki}^B = \|\xi_{k\ell}\|_2^2/d$:

$$\pi_i^r = \frac{\sum_k \alpha_{ki}^r \text{pr}_r(\mathcal{R}_{ki}^r)}{\sum_k \alpha_{ki}^r}, \quad \mathcal{R}_{ki}^A = J_{ki}, \quad \mathcal{R}_{ki}^B = h_{ki}. \quad (23)$$

Zero responses contribute zero participation. The weights are the squared loss sensitivities of the incoming and outgoing paths.

To compare the learned response with its initialization, define $\Gamma_{ki}^A = J_{ki}J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki}h_{ki}^\top$. A superscript 0 denotes the same evaluation with initial rational coefficients, at the current inputs and scales. For either role,

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2} \sqrt{\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2}}. \quad (24)$$

This is a cosine in the space of weighted kernel collections. Positive semidefiniteness and Cauchy–Schwarz give $0 \leq \kappa_i^r \leq 1$. These weighted ratios apply when their denominators are positive; otherwise the group’s rotation fractions are zero. Using the same current inputs in both collections isolates the change in the learned activation. If $\psi = \arccos \kappa$, the fraction $e = \pi(1 - \kappa^2) = \pi \sin^2 \psi$ combines the kernel angle with the breadth of its response.

For a nonzero parent and tangent, the corresponding matrix rotation is

$$a^\perp = a - \frac{\langle a, p \rangle}{\|p\|_F^2} p, \quad \tilde{q} = \sqrt{1 - e} p + \sigma \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} a^\perp, \quad (25)$$

where σ makes $\sigma \langle g, a^\perp \rangle \geq 0$. Orthogonality gives

$$\|\tilde{q}\|_F^2 = \|p\|_F^2, \quad \langle g, \tilde{q} \rangle = \sqrt{1 - e} \langle g, p \rangle + \sqrt{e} \frac{\|p\|_F}{\|a^\perp\|_F} |\langle g, a^\perp \rangle|. \quad (26)$$

The angle ϑ between p and $\tilde{q}$ therefore satisfies $\sin^2 \vartheta = \pi \sin^2 \psi$. Broad responses admit larger rotations for the same kernel departure; an unchanged kernel gives zero rotation. The common multiplier μ preserves the angle and multiplies both squared norms by μ^2 .

Attention uses the layer’s two response roles jointly. Let $\bar{\pi}_\ell^A, \bar{\pi}_\ell^B$ be group-averaged participation, and let $\kappa_\ell^A, \kappa_\ell^B$ use kernel inner products and squared norms aggregated across groups. Its rotation fraction is $e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B)$. All four role statistics lie in $[0, 1]$, so this layerwise fraction also lies in $[0, 1]$.

Table 1: TILLER training. Set $T_s = T$ for full-horizon training; otherwise use continuation optimizer $\mathcal{U}$ and state initializer $\mathcal{I}$ after T_s . Here ξ is the summed-loss derivative at a block output, scaled by the clipping factor. Probe at the first update, then periodically; reuse the latest response summaries between probes.

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- Initialize the GRAIN model, gradient moments, and response ledger.
 - 1 For $t = 1, \dots, T$, compute the minibatch loss and clipped gradient g_t .
 - 2 If $t > T_s$: initialize continuation state with $\mathcal{I}$ at $T_s + 1$, update matrices with $\mathcal{U}$, and proceed to line 11.
 - 3 On probe steps, evaluate GRAIN features h , Jacobians J , and loss-weighted participation π and reference-kernel congruence κ .
 - 4 Update gradient moments; form polar parents p and factorized adaptive directions a ; let $a^\perp = a - \langle a, p \rangle p / \|p\|_F^2$.
 - 5 For feed-forward roles set $e = \pi(1 - \kappa^2)$; for attention use $e = \sqrt{\pi^A \pi^B} (1 - \kappa^A \kappa^B)$ with layerwise summaries. Orient $a^\perp$ with the gradient. Rotate by $\tilde{q} = \sqrt{1 - e} p + \sigma \sqrt{e} \|p\|_F a^\perp / \|a^\perp\|_F$; retain p unless the candidate is finite, nondegenerate, and both directions have positive gradient contractions. Apply matrix-size scaling to obtain q .
 - 6 Form $s_i = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ and $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$.
 - 7 On probes, measure $S_{ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle$; between probes use repeated current-gradient rows at the latest measured scale. Evaluate the corresponding weight-decay responses.
 - 8 Discount the retained ledger and add current responses to obtain metric G and decay cross term r .
 - 9 Propose c in a Krylov subspace for $F(c) = -\eta_t s^\top c + \frac{\eta_t^2}{2} (c^\top G c + 2r^\top c)$ subject to $\sum_i \omega_i c_i^2 = \sum_i \omega_i$. Accept a finite proposal if it improves F , meets the norm tolerance, and both the proposal and $\mathbf{1}$ have positive layerwise gradient/momentum contractions; otherwise set $c = \mathbf{1}$.
 - 10 Apply $A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}}) A_i - \eta_t c_i q_i^A$ and the same rule to B_i ; update attention with its layer-guided q . Compress the ledger by Robust Frequent Directions.
 - 11 Update rational coefficients and remaining parameters with AdamW.
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